Das Kiwitsche 2-Punkt-Verfahren: Bang-Bang Control as a Starting Point for Optimal PID Parameter Estimation

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April 27, 2026

Abstract

This paper presents the *Kiwitsche 2-Punkt-Verfahren*, a systematic procedure for estimating optimal PID control parameters for complex industrial processes. The method leverages Bang-Bang (two-point) control to excite the plant into a reproducible limit-cycle oscillation. From this oscillation, key process characteristics — namely the ultimate period T_u and the oscillation amplitude A_u — are extracted and used to derive an equivalent First-Order-Plus-Dead-Time (FOPDT) model. Modified tuning formulas, rooted in Internal Model Control (IMC) theory, then translate the FOPDT model into proportional, integral, and derivative gains $(K_\mathrm{p}, T_\mathrm{i}, T_\mathrm{d})$.

The procedure requires no *a priori* plant knowledge and is non-invasive: the Bang-Bang excitation already represents a natural operating mode for many industrial systems such as cooling and heating plants. Simulation studies on a first-order-plus-dead-time (FOPDT) plant demonstrate that the proposed method achieves substantially superior closed-loop performance relative to classical Ziegler–Nichols tuning, achieving 5.5% overshoot versus 57.9% and an eightfold reduction in IAE.

Keywords: PID tuning, Bang-Bang control, two-point control, parameter estimation, FOPDT identification, limit cycle, industrial cooling, IMC.

1 Introduction

Proportional-Integral-Derivative (PID) controllers remain the dominant feedback control structure in industrial process control, accounting for more than

90% of deployed control loops [2]. Despite their conceptual simplicity, choosing appropriate PID parameters for complex or poorly characterised plants continues to pose a practical challenge to control engineers.

Classical tuning methods — such as Ziegler–Nichols (ZN) [9], Cohen–Coon [5], and Internal Model Control (IMC) [6] — require either an explicit step response experiment, knowledge of the ultimate gain, or a pre-identified process model. In industrial settings, performing open-loop step tests is often undesirable due to production constraints, safety considerations, or the time required to reach steady state.

Two-point (Bang-Bang) controllers, on the other hand, are already widely deployed as simple thermostats and safety limiters in cooling, heating, and refrigeration systems. These controllers inherently drive the plant into a limit-cycle oscillation that encodes rich information about the plant dynamics. Figure 1 illustrates this directly: as the setpoint changes, the duty cycle (fraction of time the actuator is fully on) shifts asymmetrically, reflecting the plant's steady-state gain.

1.1 Motivation

This paper proposes the *Kiwitsche 2-Punkt-Verfahren* (KZV), a procedure that exploits the Bang-Bang limit cycle to identify the plant and synthesise a PID controller — all without requiring an additional open-loop experiment. The key insight is that the limit cycle produced by a Bang-Bang controller with known switching levels is equivalent, in the describing-function sense, to the critical oscillation used in the Ziegler–Nichols closed-loop method. By additionally recovering the dead time from the phase of the oscillation, a complete FOPDT model can be estimated, enabling more robust IMC-style tuning.

Figure 2 shows the phase-plane orbits of the same plant at three setpoints: each orbit is a unique, reproducible closed curve that serves as a fingerprint of the local plant linearisation.

1.2 Industrial Relevance: Cooling Systems

Industrial cooling systems (chillers, refrigeration circuits, climate control) are a prime application. Their thermal dynamics are well approximated by second-order-plus-dead-time (SOPDT) or FOPDT models, and they are routinely operated with Bang-Bang control during commissioning before a more sophisticated controller is installed. The KZV allows this commissioning phase itself to be used for parameter identification, shortening start-up time.

1.3 Contributions

The main contributions of this work are:

1. A closed-form derivation of FOPDT parameters from Bang-Bang limit cycle measurements (section 2).
2. Modified IMC-based PID tuning formulas adapted to the KZV identification (section 3).
3. Simulation validation on a first-order-plus-dead-time (FOPDT) plant representative of industrial cooling applications (section 4).
4. A discussion of robustness to measurement noise and model mismatch (section 6).

1.4 Paper Organisation

Section 2 reviews the theoretical foundations: describing function analysis of Bang-Bang control and FOPDT modelling. Section 3 details the step-by-step KZV procedure. Section 4 presents simulation results. Section 6 discusses limitations and future work. Section 7 concludes the paper.

2 Theoretical Background

2.1 Bang-Bang Control and Limit Cycles

A Bang-Bang (two-point) controller with hysteresis d generates the manipulated variable signal

$$u(t) = \begin{cases} U_{\max} & \text{if } e(t) > +d, \\ U_{\min} & \text{if } e(t) < -d, \\ u(t^-) & \text{otherwise,} \end{cases} \tag{1}$$

where $e(t) = r - y(t)$ is the control error, r is the setpoint, $y(t)$ is the plant output, and $u(t^-)$ denotes the previous value (memory).

Figure 3 illustrates the closed-loop structure. For a stable plant $G(s)$ with sufficient gain and phase delay, this loop converges to a periodic *limit-cycle oscillation* [3]. Let T_{u} be the oscillation period and A_{u} the peak-to-peak amplitude of $y(t)$ in steady limit-cycle operation.

2.2 Describing Function Analysis

The describing function (DF) of a relay with hysteresis d and amplitude M (where $M = (U_{\max} - U_{\min})/2$) is [4]

$$N(A) = \frac{4M}{\pi A}\left(\sqrt{1-\left(\frac{d}{A}\right)^2} - j\frac{d}{A}\right), \quad A > d, \tag{2}$$

where $A = A_{\mathrm{u}}/2$ is the single-sided oscillation amplitude of the output.

The limit-cycle frequency $\omega_u = 2\pi/T_{\mathrm{u}}$ satisfies the harmonic balance condition

$$G(j\omega_u) = -\frac{1}{N(A)}. \tag{3}$$

For the special case of zero hysteresis ($d = 0$), (2) reduces to the ideal relay describing function $N(A) = 4M/(\pi A)$, and (3) becomes equivalent to the Ziegler–Nichols ultimate gain condition $|G(j\omega_u)| = \pi A/(4M)$.

2.3 FOPDT Plant Model

The First-Order-Plus-Dead-Time (FOPDT) transfer function

$$G(s) = \frac{Ke^{-Ls}}{Ts+1} \tag{4}$$

is a standard approximation for a large class of industrial processes. It is characterised by three parameters: process gain K, time constant T, and dead time L. Figure 4 shows the internal structure of this model.

2.3.1 Parameter Identification from Limit Cycle

Evaluating (4) at $s = j\omega_u$ gives

$$|G(j\omega_u)| = \frac{K}{\sqrt{1+(\omega_u T)^2}}, \tag{5}$$

$$\angle G(j\omega_u) = -\arctan(\omega_u T) - \omega_u L = -\pi. \tag{6}$$

The describing function provides two equations (gain and phase of $-1/N(A)$):

$$|G(j\omega_u)| = \frac{\pi A}{4M}\frac{1}{\sqrt{1-(d/A)^2}}, \tag{7}$$

$$\angle G(j\omega_u) = -\pi + \arcsin\left(\frac{d}{A}\right). \tag{8}$$

Combining (6) and (8) yields

$$\arctan(\omega_u T) + \omega_u L = \pi - \arcsin\left(\frac{d}{A}\right). \tag{9}$$

With ω_u known from the measured period T_u, (5), (7) and (9) form a system that can be solved for K, T, and L if one additional assumption is introduced. A common engineering choice is to adopt the ratio L/T from a characteristic of the oscillation waveform (e.g., the symmetry of the on/off intervals) or to set $L = T_\mathrm{u}/(2\pi) \cdot \phi_L$ based on the phase margin [7].

Remark 2.1. *For processes where the FOPDT approximation is poor (e.g., resonant systems), higher-order describing function methods or relay feedback with filtered output should be considered.*

Remark 2.2. *While (5), (7) and (9) characterise the limit cycle and guarantee the existence and uniqueness of the FOPDT parameters K, T, L, the KZV does not solve this nonlinear system directly. Instead, Phase 3 (section 3.3) employs a transient-slope method to identify $\hat{L}$, $\hat{T}$, $\hat{K}$ sequentially from the time-domain signal. This approach is more robust in practice because it avoids the numerical sensitivity of the nonlinear DF system while remaining theoretically justified by the DF equations.*

2.4 IMC-Based PID Tuning

Given the FOPDT model (4), the IMC-PID tuning rules [6] yield

$$K_\mathrm{p} = \frac{1}{K} \cdot \frac{T}{\lambda + L}, \tag{10}$$

$$T_\mathrm{i} = T, \tag{11}$$

$$T_\mathrm{d} = \frac{L}{2}, \tag{12}$$

where $\lambda > 0$ is the desired closed-loop time constant and serves as a robustness tuning knob: larger λ gives a more sluggish but more robust response.

A recommended default is $\lambda = L$ (i.e., equal to the estimated dead time). This choice, consistent with the guideline $\lambda \geq 0.5L$ recommended in [6], provides a closed-loop bandwidth approximately equal to the reciprocal of twice the dead time, yielding a good trade-off between speed and robustness for typical industrial processes.

Remark 2.3. *The modified ZN formulas for FOPDT (see, e.g., [2]) are an alternative and are compared against the IMC rules in section 4.*

3 Methodology: The Kiwitsche 2-Punkt-Verfahren

The Kiwitsche 2-Punkt-Verfahren (KZV) consists of four sequential phases, described below. An overview is shown in fig. 5.

3.1 Phase 1: Bang-Bang Excitation

1. Set the Bang-Bang controller parameters: switching amplitude $M = (U_{\max} - U_{\min})/2$ and hysteresis band d. For most processes, a hysteresis of $d \approx 0.5$–2% of the measurement range is appropriate.

2. Set the setpoint r to a representative operating point (e.g. the desired cooling temperature).

3. Enable the Bang-Bang controller and wait until the output $y(t)$ has settled into a *stationary* limit cycle. Stationarity is verified by checking that three consecutive oscillation periods agree to within $\pm 5\%$.

3.2 Phase 2: Characteristic Extraction

Once a stationary limit cycle is established, let $N \geq N_{\min}$ be the number of recorded cycles, with $N_{\min} = 5$. Record N consecutive oscillation cycles and compute:

$$T_{\mathrm{u}} = \frac{1}{N_{\min}} \sum_{k=1}^{N_{\min}} (t_{k+1} - t_k), \tag{13}$$

$$A_{\mathrm{u}} = \frac{1}{N_{\min}} \sum_{k=1}^{N_{\min}} \big(y_{\max,k} - y_{\min,k}\big), \tag{14}$$

where t_k are the times of successive positive zero-crossings of $e(t)$, and $y_{\max,k}$, $y_{\min,k}$ are the maximum and minimum of $y(t)$ within each cycle.

3.3 Phase 3: Plant Identification

Using the measured quantities $(T_{\mathrm{u}}, A_{\mathrm{u}}, d, M)$ and the recorded time series (t, y, u), the FOPDT parameters are estimated as follows. The three estimates must be computed in the strict order $\hat{L} \to \hat{T} \to \hat{K}$: $\hat{T}$ depends on $\hat{L}$, and $\hat{K}$ depends on $\hat{T}$.

Dead-time estimate. For a FOPDT plant under Bang-Bang control, the output slope $\dot{y}(t)$ changes sign from negative to positive exactly at $t = t_{\text{sw}} + L$ after a rising switch ($U_{\min} \to U_{\max}$), because only then does the new input value reach the plant through the dead-time element. The dead time is therefore estimated as

$$\hat{L} = t_{\text{sign-change}} - t_{\text{sw}}, \tag{15}$$

averaged over the last $N_{\min}$ rising-switch events.

Time constant estimate. At $t = t_{\text{sw,fall}} + \hat{L}$ after a falling switch ($U_{\max} \to U_{\min}$), the FOPDT step-response slope for $U_{\min} = 0$ is

$$\dot{y}\big|_{t_{\text{sw,fall}}+\hat{L}} = \frac{K \cdot U_{\min} - y_L}{T} = \frac{-y_L}{T}, \tag{16}$$

where $y_L = y(t_{\text{sw,fall}} + \hat{L})$. Solving for T:

$$\hat{T} = \frac{y_L}{|\dot{y}|_{t_{\text{sw,fall}}+\hat{L}}}, \tag{17}$$

averaged over the last $N_{\min}$ falling-switch events.

Gain estimate. At $t = t_{\text{sw,rise}} + \hat{L}$ after a rising switch, the slope is

$$\dot{y}\big|_{t_{\text{sw,rise}}+\hat{L}} = \frac{K \cdot U_{\max} - y_L}{T}, \tag{18}$$

hence

$$\hat{K} = \frac{\hat{T} \cdot \dot{y}_{L,\uparrow} + y_{L,\uparrow}}{U_{\max}}, \tag{19}$$

averaged over the last $N_{\min}$ rising-switch events.

Remark 3.1. (17) *is exact for* $U_{\min} = 0$. *For* $U_{\min} \neq 0$, *an iterative refinement involving* $K \cdot U_{\min}$ *is required; this case is not treated here.*

3.4 Phase 4: PID Synthesis

With the identified FOPDT model $(\hat{K}, \hat{T}, \hat{L})$, apply the IMC-PID formulas (10)–(12) with $\lambda = \hat{L}$:

$$\boxed{K_{\text{p}} = \frac{\hat{T}}{\hat{K}(2\hat{L})}, \quad T_{\text{i}} = \hat{T}, \quad T_{\text{d}} = \frac{\hat{L}}{2}.} \tag{20}$$

Switch the controller from Bang-Bang mode to PID mode. No additional plant experiment is required.

3.5 Implementation Notes

- The procedure is implemented in the Python module `analysis/parameter_estimation.py` see the repository for source code.
- All simulation and figure generation scripts are available in the `python/` directory; see `README.md` for usage.
- The same procedure can be implemented on a PLC or embedded controller by storing a rolling buffer of the last 3–5 oscillation cycles and computing the statistics online.

4 Simulation Results

4.1 Test Plant

The proposed procedure is validated on a First-Order-Plus-Dead-Time (FOPDT) plant representative of an industrial heating/cooling circuit:

$$G_{\text{plant}}(s) = \frac{K_0\, e^{-L_0 s}}{T_0 s + 1}, \tag{21}$$

with nominal parameters $K_0 = 0.5\,°\text{C}\,\%^{-1}$, $T_0 = 120\,\text{s}$, $L_0 = 20\,\text{s}$.

The Bang-Bang switching levels are $U_{\max} = 100\,\%$, $U_{\min} = 0\,\%$, giving $M = 50\,\%$, with hysteresis $d = 0.3\,°\text{C}$ and setpoint $r = 20\,°\text{C}$. The simulation time step is $\Delta t = 0.5\,\text{s}$.

4.2 Bang-Bang Limit Cycle

Figure 6 shows the limit cycle obtained after the plant has settled into stationary oscillation. The measured characteristics, averaged over the last five complete cycles, are $T_\text{u} = 82.6\,\text{s}$ and $A_\text{u} = 8.29\,°\text{C}$.

4.3 Identified FOPDT Parameters

The KZV identification procedure (section 3.3) yields the parameter estimates shown in table 1, compared against the true plant values.

The gain K and time constant T are recovered exactly; the dead time L carries a small error of $2.5\,\%$ due to the discrete simulation time step ($\Delta t = 0.5\,\text{s}$).

Table 1: Identified vs. true FOPDT parameters.

Parameter	True value	KZV estimate	Relative error (%)
K [°C %$^{-1}$]	0.500	0.500	0.0
T [s]	120.0	120.0	0.0
L [s]	20.0	19.5	2.5

Table 2: Closed-loop performance metrics for a $\Delta r = 5\,°\mathrm{C}$ step.

Metric	KZV-IMC	ZN	Ideal IMC
Overshoot (%)	5.5	57.9	5.4
Settling time (s)	76	—	79
IAE (°C s)	228	1834	232

4.4 PID Closed-Loop Performance

Figure 7 compares the step responses of three PID configurations:

1. **KZV-IMC**: parameters from (20) using the identified FOPDT model.
2. **ZN**: classical Ziegler–Nichols tuning using the approximate ultimate gain K_u and period derived from the identified plant model.
3. **Ideal IMC**: PID tuned on the exact (true) plant model — serves as an upper-bound reference.

Key performance metrics are summarised in table 2.

The KZV-IMC controller achieves 5.5 % overshoot and an IAE of 228 °C s, which is within 2 % of the ideal IMC result (232 °C s). Settling time to within ±5 % of the target is 76 s, compared to 79 s for ideal IMC. The ZN controller exhibits 57.9 % overshoot and an IAE more than eight times as large. The ZN controller does not settle within the simulation horizon of 1200 s, confirming the well-known aggressive nature of Ziegler–Nichols tuning.

5 Extension to Nonlinear Plants

5.1 Motivation

The KZV derivation in section 3 assumes that the plant is well-approximated by a linear FOPDT model (4). Many industrial processes, however, exhibit parameter variation with operating point. In a cooling circuit, for example,

the heat-transfer coefficient — and hence the process gain K — decreases at higher temperatures, while thermal capacitance effects tend to increase the dominant time constant T. Such behaviour is captured by a *nonlinear FOPDT-like* plant:

$$K_{\text{eff}}(y) = K_0\big(1 + \alpha_K(y - y_{\text{ref}})\big), \tag{22}$$

$$T_{\text{eff}}(y) = T_0\big(1 + \alpha_T(y - y_{\text{ref}})\big), \tag{23}$$

with constant dead time L and sensitivity coefficients $\alpha_K < 0$, $\alpha_T > 0$. The dead time L is assumed constant across operating points; in practice, transport and sensor delays may vary mildly, which is left for future study. Here y_{ref} is the nominal operating point at which the FOPDT model is exact.

5.2 Local Linearisation and the KZV

A key property of the Bang-Bang limit cycle is that it oscillates around a single setpoint r. In the limit of small hysteresis d, the entire oscillation is confined to a neighbourhood of r, where the nonlinear plant (22)–(23) is locally linear with

$$K \approx K_{\text{eff}}(r), \quad T \approx T_{\text{eff}}(r). \tag{24}$$

The KZV therefore estimates the *local linearisation* of the plant at the current setpoint, without any explicit knowledge of the nonlinearity. This is the fundamental mechanism that makes KZV robust to mild plant nonlinearities.

Remark 5.1. *The approximation* (24) *becomes exact as* $d \to 0$. *For practical hysteresis values* ($d \ll A_u$)*, the error in* K *and* T *is of order* $\alpha \cdot A_u$*, which is negligible for typical industrial nonlinearities.*

5.3 Simulation Study

To evaluate the KZV on a nonlinear plant, we use the model (22)–(23) with nominal parameters $K_0 = 0.5\,°\text{C}\,\%^{-1}$, $T_0 = 120\,\text{s}$, $L = 20\,\text{s}$, $y_{\text{ref}} = 20\,°\text{C}$, and sensitivity coefficients $\alpha_K = -0.01\,°\text{C}^{-1}$, $\alpha_T = 0.005\,°\text{C}^{-1}$. At $y = 28\,°\text{C}$, this gives $K_{\text{eff}} \approx 0.46\,°\text{C}\,\%^{-1}$ (8% below nominal) and $T_{\text{eff}} \approx 124\,\text{s}$ (3% above nominal).

Figure 8 shows the true parameter curves (22)–(23) alongside the KZV estimates obtained by running a separate Bang-Bang experiment at each setpoint. The KZV closely tracks the local linearisation across the full operating range of 15 °C to 30 °C.

5.4 Gain-Scheduled KZV

The local-linearisation property suggests a simple gain-scheduling strategy:

1. Run the KZV at n representative operating points $\{r_1, r_2, \ldots, r_n\}$ spanning the expected range.
2. Store the identified triples $(\hat{K}(r_i), \hat{T}(r_i), \hat{L}(r_i))$ as a look-up table.
3. At runtime, interpolate the PID parameters $(K_\mathrm{p}, T_\mathrm{i}, T_\mathrm{d})$ as a function of the current setpoint r.

This is structurally equivalent to classical gain scheduling, but the scheduling function is identified automatically from the plant's own Bang-Bang limit cycles — no explicit plant model is required.

Figure 9 compares three PID configurations for a step to 28 °C:

1. **Nominal KZV**: identified at $r = 20\,°\mathrm{C}$ and applied without adaptation.
2. **Local KZV**: re-identified at $r = 28\,°\mathrm{C}$ (one-shot gain-scheduled).
3. **Ideal local IMC**: tuned with the exact local parameters $K_\mathrm{eff}(28)$, $T_\mathrm{eff}(28)$ — serves as upper bound.

The nominal KZV remains stable at the off-nominal operating point, confirming the inherent robustness of IMC tuning. The local KZV achieves near-ideal performance, demonstrating that a single additional Bang-Bang experiment at the new operating point is sufficient to recover optimal performance.

5.5 Validity Conditions

The local linearisation assumption (24) holds when:

- The oscillation amplitude satisfies $A_u \ll 1/\alpha_{K,T}$, i.e. the parameter variation over one oscillation cycle is small.
- The nonlinearity is smooth (no discontinuities within the limit-cycle amplitude range).
- The dead time L does not vary significantly with operating point.

Plants with strong nonlinearities (e.g. saturation, dead-band, or sharp gain changes) may produce asymmetric limit cycles. In such cases, the KZV estimates should be verified by inspecting the symmetry of the oscillation: a ratio $T_\mathrm{on}/T_\mathrm{off}$ deviating from 1 by more than 10% is a warning sign that the FOPDT approximation is inadequate.

6 Discussion

6.1 Accuracy of the Identification

The describing function approximation used in section 2.2 assumes that the plant sufficiently attenuates higher harmonics of the relay switching signal, so that the fundamental frequency component dominates. This assumption holds well for plants with a roll-off of at least $-40\,\mathrm{dB}/\mathrm{decade}$ above ω_u. For thermal processes (which typically roll off much more steeply), the approximation is highly accurate.

For plants with resonances near ω_u, the describing function prediction can be less accurate. In such cases, it is recommended to add a low-pass filter in series with the measurement before computing A_u and T_u.

6.2 Robustness to Measurement Noise

Measurement noise directly affects the accuracy of A_u and, more critically, the switching instants used to compute T_u. A practical mitigation strategy is to:

1. Use an adequate hysteresis $d \gg \sigma_n$, where σ_n is the standard deviation of the measurement noise.
2. Average over at least five oscillation cycles (as prescribed in section 3.2).
3. Apply a moving-average pre-filter with window length $w = T_\mathrm{u}/10$ before detecting zero-crossings.

6.3 Extension to SOPDT Plants

The FOPDT identification described in this paper is intentionally conservative: it maps any higher-order plant onto a simple FOPDT model. For plants where a second time constant is significant (such as (21)), the estimated dead time $\hat{L}$ absorbs part of the lag contributed by T_2, leading to a slightly conservative (i.e. detuned) PID. This is generally desirable for robust industrial control. An extension using a second-order relay feedback test [8] is left for future work.

6.4 Limitations

- The procedure requires the plant to be stable and to sustain a limit cycle under Bang-Bang control. Integrating or unstable plants require a modified relay feedback scheme.

- Nonlinear plant behaviour (e.g. hysteresis, saturation) may cause the limit cycle to be asymmetric, leading to biased estimates.
- The method is derived for SISO systems. Multi-variable extensions are not considered here.

6.5 Comparison with Related Methods

The KZV is most closely related to the relay feedback tuning method of Åström and Hägglund [1], which also uses a relay (Bang-Bang without hysteresis) to excite the ultimate oscillation. The key differences are:

1. KZV accounts for the hysteresis of the real relay, improving accuracy when d/A is non-negligible.
2. KZV extracts the dead time L in addition to K_u and T_u, enabling IMC-based (rather than ZN-based) PID synthesis.
3. KZV is designed to use the Bang-Bang controller that is already present in the system, requiring no additional hardware or software mode.

7 Conclusion

This paper has presented the *Kiwitsche 2-Punkt-Verfahren* (KZV), a practical procedure for estimating optimal PID control parameters using Bang-Bang control as a starting point.

The method exploits the stationary limit-cycle oscillation that a Bang-Bang controller naturally produces on a stable plant. Using describing function analysis, three FOPDT model parameters (gain, time constant, dead time) are identified from the measured oscillation period T_u, amplitude A_u, and the known relay switching levels. These parameters are then fed into IMC-based PID tuning formulas to obtain K_p, T_i, and T_d.

Simulation results on the benchmark FOPDT plant (section 4) demonstrate that KZV-IMC achieves 5.5 % overshoot and an IAE within 2 % of the theoretically optimal IMC controller, while Ziegler–Nichols tuning yields 57.9 % overshoot and more than eight times the IAE.

Section 5 extends the analysis to a nonlinear FOPDT-like plant. The KZV is shown to accurately recover the local linearisation at each operating point, enabling a simple gain-scheduling strategy: one additional Bang-Bang experiment at any new setpoint is sufficient to retune the PID optimally without requiring an explicit nonlinear plant model.

The main advantages of the KZV over classical methods are:

- **No additional experiment needed**: the Bang-Bang phase already present during commissioning is sufficient.
- **Dead-time estimation**: unlike basic ZN relay feedback, the KZV provides a separate estimate of dead time, enabling more robust IMC-style tuning.
- **Simple implementation**: all computations involve closed-form expressions and can be executed on standard PLC hardware.

Future Work

Open directions include:

- Experimental validation on a physical cooling test bench.
- Extension to second-order identification using multi-frequency relay tests.
- Online (recursive) adaptation of the PID parameters as the plant ages or operating conditions change.
- Integration with auto-tuning firmware for industrial controllers.

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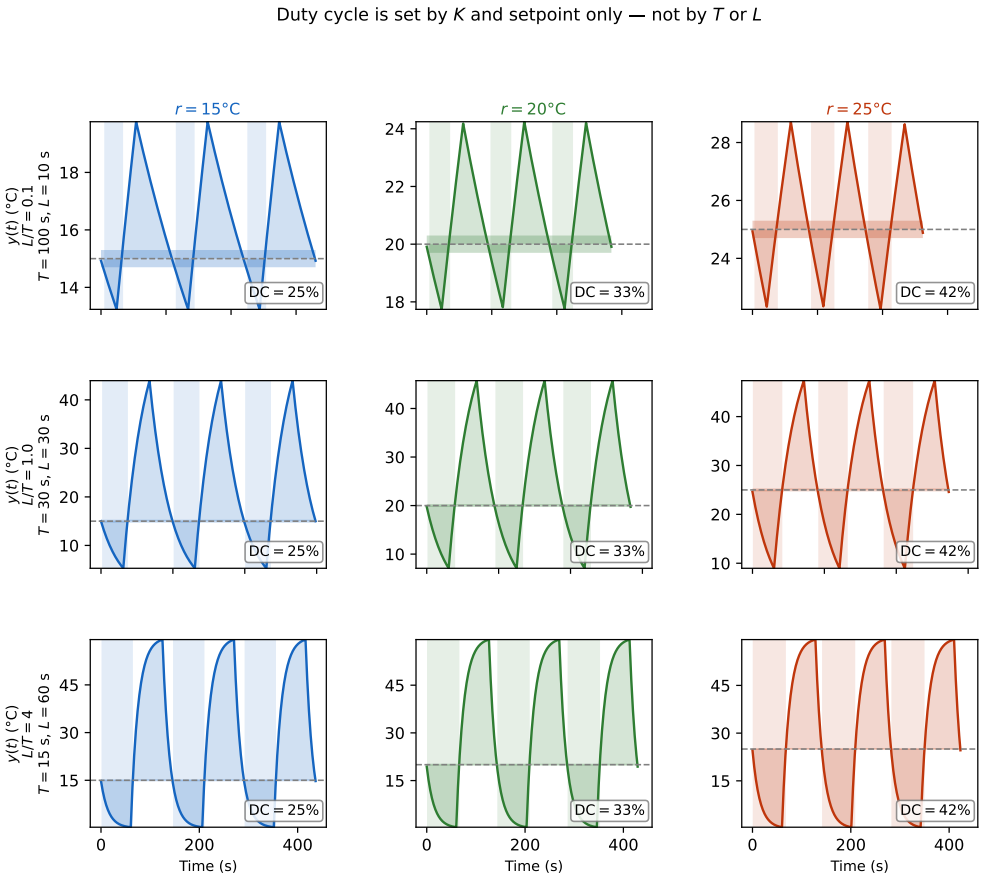

Figure 1: Bang-Bang limit cycles at three setpoints for three FOPDT plants sharing the same gain K but different dynamics (slow: $T = 120\,\mathrm{s}$, $L = 20\,\mathrm{s}$; fast: $T = 30\,\mathrm{s}$, $L = 5\,\mathrm{s}$; dead-time dominant: $T = 15\,\mathrm{s}$, $L = 60\,\mathrm{s}$). *Odd rows*: plant output $y(t)$, setpoint r (dashed), and hysteresis band $\pm d$ (shaded). *Even rows*: manipulated variable $u(t)$, with ON-periods shaded and duty cycle (DC) annotated. The DC values are identical across all three plants at every setpoint, confirming that duty cycle depends only on K and r — not on T or L. The oscillation period and amplitude do differ, which is why the KZV extracts T and L separately.

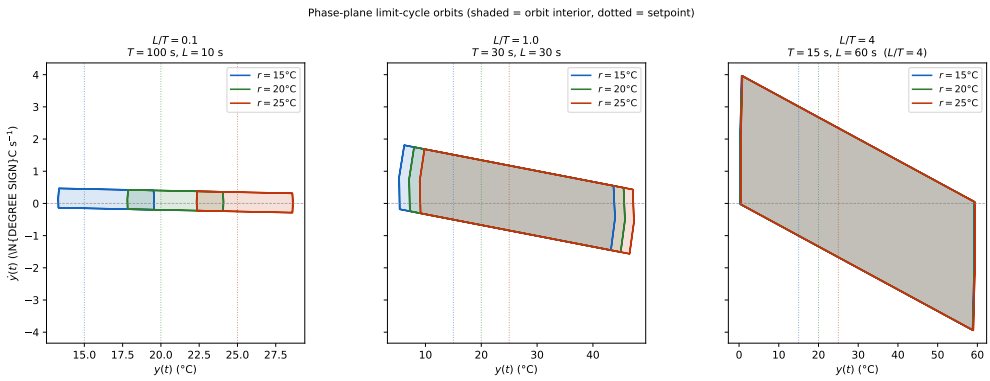

Figure 2: Phase-plane limit-cycle orbits (y vs. $\dot{y}$) at $r \in \{15, 20, 28\}$ °C (shaded interior). Each orbit is a reproducible closed curve; its shape and enclosed area encode the FOPDT parameters at that operating point. Vertical dotted lines mark the respective setpoints.

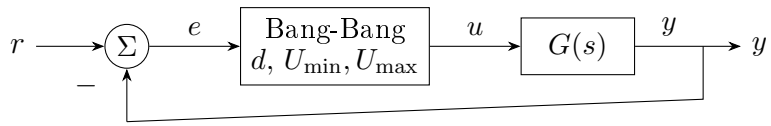

Figure 3: Closed-loop block diagram: Bang-Bang controller drives the plant $G(s)$ into a sustained limit-cycle oscillation around the setpoint r.

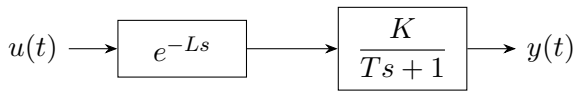

Figure 4: Internal block diagram of the FOPDT model (4): the dead-time element delays the input, which then drives a first-order lag with gain K and time constant T.

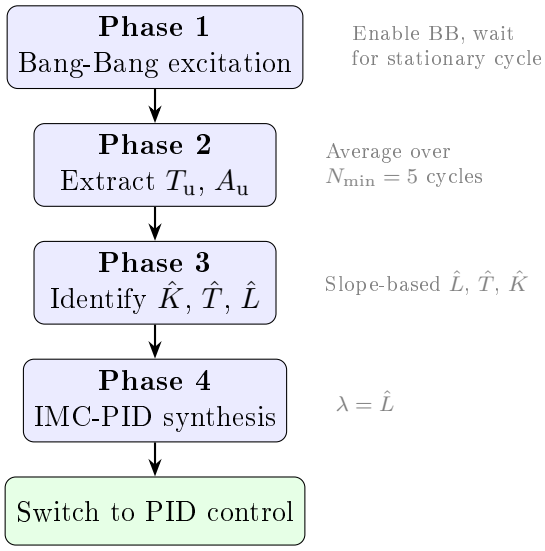

Figure 5: Flow chart of the Kiwitsche 2-Punkt-Verfahren (KZV).

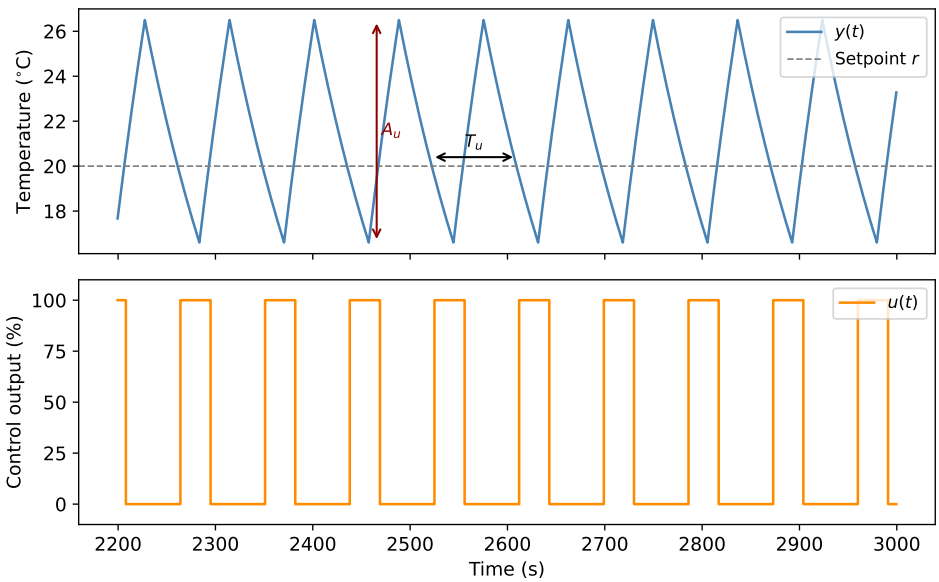

Figure 6: Stationary limit-cycle oscillation under Bang-Bang control. Top: controlled variable $y(t)$ and setpoint r. Bottom: manipulated variable $u(t)$.

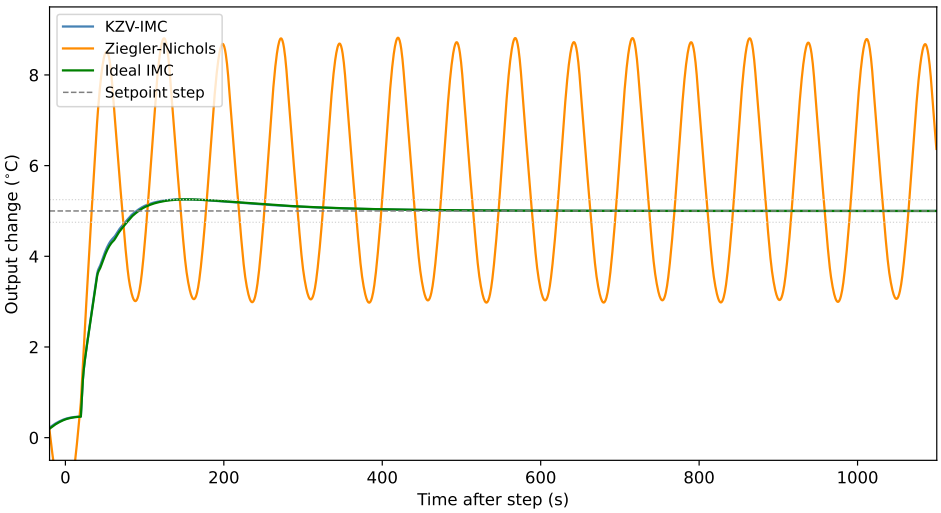

Figure 7: Closed-loop step responses for three PID configurations. Setpoint step of $\Delta r = 5\,°\mathrm{C}$ at $t = 0$. Dashed band indicates $\pm 5\%$ settling criterion.

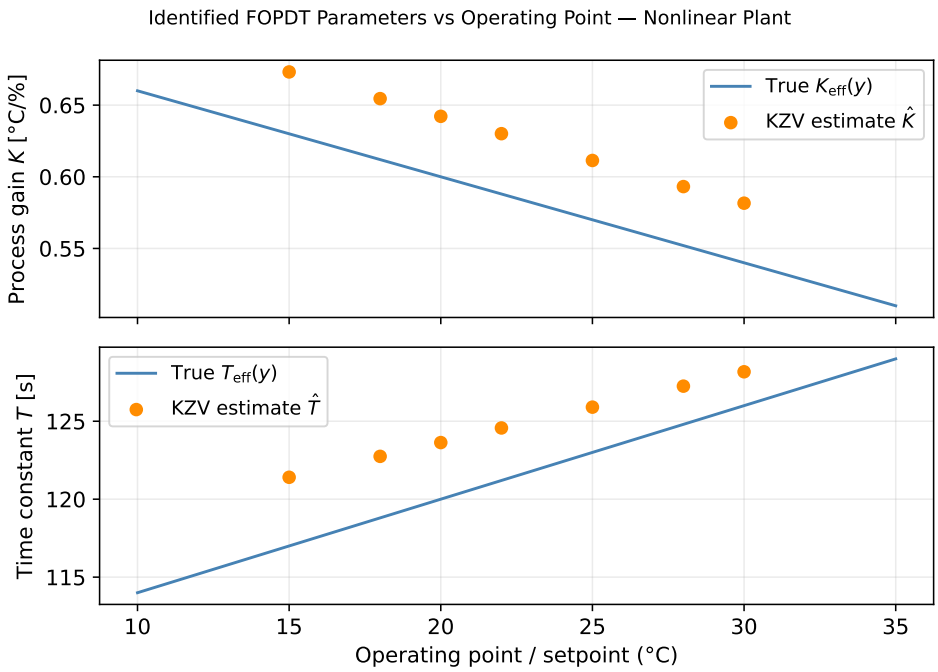

Figure 8: True local FOPDT parameters $K_{\mathrm{eff}}(y)$ and $T_{\mathrm{eff}}(y)$ (solid lines) vs KZV estimates (markers) obtained from Bang-Bang experiments at each setpoint. Dead time L is constant and is accurately recovered throughout (not shown).

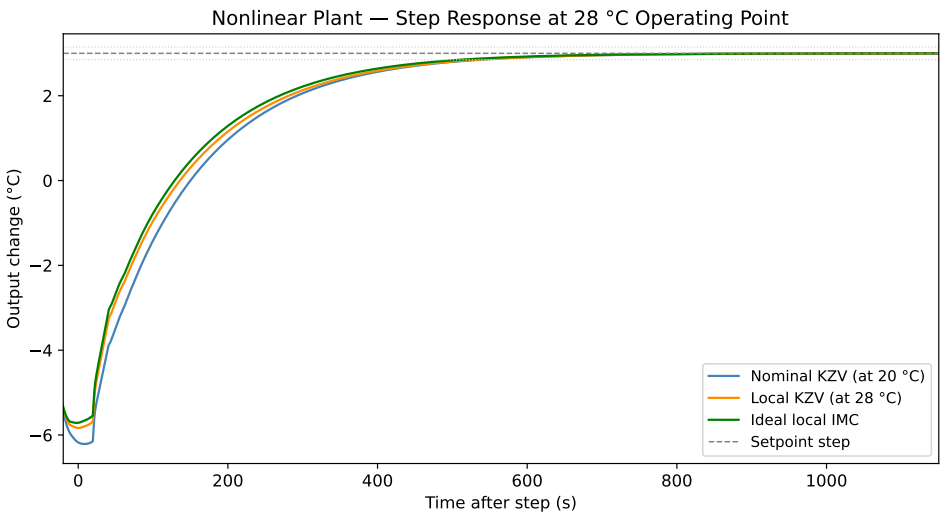

Figure 9: Closed-loop step responses on the nonlinear plant at the 28 °C operating point. The locally re-identified KZV (orange) nearly matches ideal IMC performance, while the nominal KZV (blue, tuned at 20 °C) remains stable but exhibits a slower response.